

The Importance of Dark Web Access in Cyber Threat Intelligence

Victor Obojo

*Cyber Threat Intelligence Division
CyberDome Nigeria
Abuja, Nigeria
victor.o@cyberdome.net
0009-0005-3450-4906*

Richard A. Ikuesan

*Computing and Applied Technology
College of Technological Innovation, Zayed University
Abu Dhabi, UAE
richard.ikuesan@zu.ac.ae*

AbdulKabir Adekanye

*Cyber Threat Intelligence Division
CyberDome Nigeria
Abuja, Nigeria
adekanyekabir@gmail.com*

Elias Iortom

*Cyber Threat Intelligence Division
CyberDome Nigeria
Abuja, Nigeria
elias.i@cyberdome.net*

Abstract—Cyber Threat Intelligence (CTI) is essential in informing proactive cyber defence, especially in Nigeria’s most significant sectors: financial services, telecommunications, and healthcare. This study reviews the current body of CTI literature, examines intelligence obtained from the Dark Web, and determines impending threats, including Ransomware-as-a-Service, zero-day exploits, and phishing toolkits aimed at these sectors. Performing qualitative platform analysis of the likes of RaidForums and Exploit.in, we determine sectoral vulnerabilities and offer pragmatic suggestions on developing Nigeria’s cyber defence system. The findings underscore the Dark Web’s role as a vital source of intelligence to counter cyber threats under low-resource conditions, filling Nigeria’s gaps in CTI capabilities.

Index Terms—dark web, cti, cybersecurity, cybercrime, threat analysis

I. INTRODUCTION

In today’s hyperconnected digital age, Cyber Threat Intelligence (CTI), the practice of collecting and assessing information to anticipate and combat cyber threats, is now vital to safeguarding organizations and nations [1]. The stakes cannot be higher for Nigeria, which is digitizing at a frenetic pace. With an internet penetration of 51% as of 2022, millions of Nigerians have embraced mobile banking, e-commerce, and other digital services, powering economic development. But this accelerated growth also puts the country increasingly in harm’s way of advanced cyber threats like ransomware, data breaches, and identity theft, so effective CTI is an urgent necessity [2].

One of the principal causes of such threats is the Dark Web, an unseen component of the internet that can only be accessed using special tools like Tor, in which anonymity conceals illegal activities. Far more than a seedy corner of the web, the Dark Web is a thriving marketplace for threat actors [3], where everything from pilfered credentials to ransomware kits and

zero-day exploits tools that can take down systems within a matter of seconds are for sale [4]. Forums like RaidForums and Exploit.in, for instance, routinely host transactions involving Nigerian data, including compromised bank accounts and phishing templates targeting local users [5].

Additionally, in July 2023, Globacom Nigeria, a leading telecommunications company, suffered a ransomware attack by the ALPHV group. The attackers claimed to have been in control of the network for 12 days undetected and demanded a ransom of \$2.5 million. The incident underscores the tangible risks of ransomware attacks facilitated through toolkits purchased on the Dark Web. This study examines how the integration of Dark Web intelligence into CTI frameworks [6] can transform Nigeria’s cyber threat fighting potential. By monitoring such underground forums, we learn about trends Ransomware-as-a-Service (RaaS) and the sale of region-specific exploits, for instance, that predict future threats [4]. Our research demonstrates that proactive Dark Web analytics enhance threat discovery and allow Nigeria to stay ahead of its adversaries in an increasingly dangerous digital age.

II. LITERATURE REVIEW

Cyber Threat Intelligence (CTI) has become vital in implementing effective cybersecurity strategies globally. According to [7], CTI is the systematic collection, analysis, and application of information about potential or actual cyber threats in improving security within an organization. In highly sensitive sectors, CTI plays a vital part in the prevention of data breaches, service disruptions, and financial loss [8]. For Nigeria, with its rapidly accelerating digital transformation, CTI is very important in addressing the unique challenges that cyber threats present in a developing economy [2].

A. The Role in Cyber Threat Intelligence

The Dark Web is a part of the internet that is intentionally hidden and accessible only with special tools, like Tor. It is unindexed by conventional search engines, which makes it a haven for many illegal activities, including the sale of stolen data, ransomware kits, and hacking tools. According to S [9], the Dark Web offers critical information on evolving threats, hostile tactics, and vulnerabilities, which makes it indispensable to Cyber Threat Intelligence (CTI) efforts.

This thus makes Exploit.in and RaidForums among the most visible hotspots for Nigerian industries. For instance, RaidForums has traditionally listed stolen credentials from Nigerian banks, while Exploit.in hosts zero-day vulnerabilities targeting telecommunications infrastructure. It showcases the diverse and dynamic nature of threats encountered on the Dark Web [10].

B. Sector-Specific Threats on the Dark Web Relevant to Nigeria

There are unique challenges to cybersecurity in the Nigerian space, specifically for financial services, telecommunications, and healthcare. For example, the banking sector is generally always at the centre of phishing attacks and selling stolen credentials on platforms such as RaidForums. Equally, telecommunications companies face zero-day vulnerabilities, bought and traded on Exploit.in, which allow hackers to breach mobile networks and intercept calls [11] [12]. Table I sheds more light on attacks targeting specific sectors.

The healthcare sector faces risks from ransomware attacks, where attackers use ransomware tools obtained from the Dark Web to encrypt critical medical records. This vulnerability is exacerbated by the reliance on legacy systems [13], making healthcare institutions lucrative targets for cybercriminals on the Dark web [14].

C. Challenges and Ethical Considerations

The legal and ethical issues involved in monitoring the Dark Web are enormous. The fine line between intelligence gathering and illegitimate activities has to be taken care of by any organization. Ethical Dark Web monitoring involves passive observation techniques, strong anonymization, and a strong adherence to laws on cybersecurity as observed by [15]. In developing nations like Burkina Faso, Lesotho, Malawi, Djibouti, Nigeria, Mozambique etc. the limitation in resources and access to sophisticated monitoring tools presents more challenges in the integration of Dark Web intelligence into CTI programs.

D. Gaps in Existing Research

Although there are numerous Cyber Threat Intelligence (CTI) studies all over the globe, there are astonishing gaps in capturing the unique challenges for industries in developing nations. Most of the literature focuses majorly on developed economies, often at the expense of resource constraints, technology disparities, and specific vulnerabilities typical of developing nations like Nigeria [11] [16]. For instance, research

traditionally looks at sophisticated cybersecurity systems for nations with established digital infrastructures and thus do not offer much guidance for nations struggling with antiquated systems and limited budgets [17].

Additionally, regional threats also manifest in trends much dissimilar to those observed at a global level. As an illustration, Nigerian bank-focused phishing kits have been found on platforms such as RaidForums, showing concentrated cybercrime against the financial sector of the nation. Ransomware assaults against unpatched vulnerabilities within health systems are similarly a source of worry but these phenomena don't find prominent mention in international CTI reports. Industry-based threats such as these need sophisticated measures that lack in large-scale studies.

The second area overlooked is the provision of tools and resources for Dark Web monitoring. The majority of studies concentrate on using sophisticated, expensive CTI tools that are typically out of reach for organizations in developing countries. This inability to make realistic recommendations further widens the gap between global research and its applicability in countries like Nigeria.

This research hopes to bridge such gaps by offering localized knowledge on Dark Web threat propagation and making practical recommendations tailored to Nigeria's main economic sectors. By addressing such tailored vulnerabilities and deficits, this research helps the creation of more comprehensive and effective cybersecurity controls.

III. METHODOLOGY

This study follows a qualitative method in studying and analyzing Dark Web cyber threats targeting Nigeria's most important economic sectors. This kind of method is appropriate for the study of complex and dynamic phenomena like cyber threats, which require deep contextual understanding and interpretive analysis rather than linear statistical or numerical evaluation. Besides, a systematic dataset of Dark Web intelligence monitoring is available here¹. By emphasis on unstructured data analysis and contextual threats importance, this study provides a comprehensive examination of the vulnerabilities and mitigation processes following particular industries. Additionally, the approach enables an in-depth exploration of the behaviors, trends, and activities manifested on Dark Web sites, which are challenging to measure based on quantifiable metrics.

A. Dark Web Access and Security Setup

The dark web can only be securely accessed through the use of sophisticated tools and a complex security framework to provide anonymity and protection for the researcher. In this study, the Tor browser was utilized to facilitate encrypted access to '.onion' websites, while virtual machines created isolated research environments that effectively minimized the risk of system compromise. Virtual Private Networks (VPNs) and firewalls were also utilized to obscure internet traffic and

¹Dataset URL: <https://github.com/V-Obojo/Dark-web-Threat-monitoring-dataset>

TABLE I: Sector-Specific Threats from the Dark Web for Nigeria’s Critical Industries

Sector	Dark Web Platform	Threat Type	Data Collected
Financial Institutions	Various Dark Web Platforms	Stolen Credentials, Phishing Kits, Ransomware-as-a-Service (RaaS)	Bank account details, MFA bypass kits, customized ransomware for financial networks
Telecommunications	Various Dark Web Platforms	Zero-Day Exploits, DDoS Services, Botnets	Network vulnerabilities, denial-of-service tools, botnet rentals targeting telecom networks
Healthcare	Dark Web Marketplaces, Ransomware Platforms	Ransomware, Stolen Patient Data	Healthcare ransomware kits, patient record sale listings, and network exploits for healthcare systems
Electricity and Gas Supply	Black Market Forums	DDoS Attacks, Malware for Infrastructure Disruption	Botnet services, malware for control systems, ransomware targeting utility providers
Chemical and Pharmaceutical Products	Dark Web Shops, Exploit.in	Intellectual Property Theft, Industrial Espionage	Stolen patents, chemical formulas, and proprietary manufacturing methods
Water Supply and Waste Management	Black Market Forums, Exploit.in	System Vulnerabilities, Ransomware	Infrastructure network vulnerabilities, ransomware for water purification plants
Air Transport	Dark Web Forums	Infrastructure Attacks, Credential Theft	Airline system exploits, pilot and passenger data sales
Water Transport	Dark Web Forums	Operational Disruptions, Network Exploits	Exploits for maritime logistics, shipping schedules, and transportation networks
Information and Communication	Exploit.in, DarkOwl	Data Theft, Ransomware	Hacking tools, communication platform vulnerabilities, sensitive data sales
Metal Ores	Specialized Marketplaces	Espionage, Data Breaches	Industrial secrets, exploration data, export license details
Note: The dark web platforms referenced in this table include, but are not limited to, RaidForums, Exploit.in, various ransomware markets, and other specialized dark web marketplaces.			

protect access points. These measures ensured the integrity of the research approach while guarding vulnerable systems against potential hazards throughout the information-gathering process.

B. Data Source Selection and Justification

Given the nature of the Dark Web, the study prioritizes high-traffic forums and marketplaces related to Nigeria’s most important economic sectors. Websites such as RaidForums, Exploit.in, and Genesis Market were selected as they were reputed for selling stolen credentials, ransomware kits, and zero-day exploits.

- **RaidForums:** It is renowned for ads of illicit financial information, such as Nigerian bank account details, which makes it essential for tracking threats to the financial industry.
- **Exploit.in:** Routinely has discussions of zero-day exploits on enterprise platforms, which have direct application to vulnerabilities within Nigeria’s telecommunication and utility sectors.
- **Genesis Market:** Focuses on the sale of digital fingerprints and breached account access, threatening various industries, such as healthcare and government services.

The choice of these platforms was guided by industry reports generated by Europol and Trend Micro, thereby ensuring their applicability to the specified industries and alignment with the research aim of finding threats specific to these industries.

C. Data Collection Tools and Techniques

A hybrid strategy that merged automated tools and human processes was employed to collect comprehensive intelligence from the Dark Web.

- **Automated Monitoring:** Software like DarkOwl and Recorded Future was set up to constantly scan for keywords such as “Nigerian banks” and “ransomware kits” so that real-time data could be collected from various sources. This facilitates the detection of emerging threats in Nigeria’s economy.
- **Web Scraping:** Custom web scraping bots systematically collected data from popular forums at regular time intervals.
- **Manual Validation:** Security experts validated valuable posts to check the validity and applicability of the collected information.

The integration of these approaches enabled the accurate identification of emerging threats across Nigeria’s economic sectors, thus directly supporting the research aim of analyzing vulnerabilities particular to each sector.

1) *Data Validation and Filtering:* Data validation was a crucial step to guarantee intelligence gathered was of high integrity. To reduce false positives and counter misinformation, all data gathered was methodically cross-referenced with publicly available cybersecurity reports from trusted sources, including Kaspersky and Trend Micro. The data that was validated was then classified into threat classes (e.g., ransomware, phishing) through Threat Intelligence Platforms (TIPs) such as MISP and Anomali ThreatStream. These platforms also enabled the discovery of Indicators of Compromise (IOCs), such as malware signatures and malicious IP addresses, thereby enhancing the validity of the intelligence used in the research.

D. Threat Analysis and Categorization

After validation, the information was classified into thematic categories by targeted sectors and threat type. For example:

- 1) **Financial Services:** Breached authentication data and phishing attacks are the focus.

- 2) **Telecommunications:** Scanning for zero-day vulnerabilities of network infrastructure.
- 3) **Healthcare:** Analysis of ransomware attacks on patient data management systems.

This grouping allowed for a detailed, sectoral analysis of threats in support of the research aim of delivering customized intelligence for Nigerian organizations.

E. Legal and Ethical Considerations

Due to the illicit nature of most Dark Web activities, this research rigorously followed passive observation only, refraining from active participation in any illegal activities or interaction with threat actors. No data was bought, and all intelligence was obtained from publicly available forums and marketplaces. Personally identifiable information was anonymized or omitted to avoid privacy infringement.

The research followed national regulations, such as Nigeria's Cybercrime Act 2015 and international cybercrime legislations, ensuring ethical integrity throughout the research process.

F. Constraints and Obstacles

The ever-changing nature of the Dark Web was challenging, as both forums and marketplaces tended to alter URLs or go offline frequently. To address this issue, redundancy of monitoring was used across multiple sources by the study. Yet, resource and legal limitations restricted interaction with specific threat actors directly, potentially impacting the exhaustiveness of some of the results. Despite these difficulties, the multi-dimensional strategy facilitated an extensive examination of existing data.

IV. FINDINGS

This section discusses the lessons learned from Dark Web monitoring and lists the results by emerging threats, sector-specific vulnerabilities, and real-life case studies.¹ The results are organized into three primary categories: Emerging Threats, Sector-Specific Vulnerabilities, and Case Studies that illustrate real-world implications of the Dark Web intelligence.

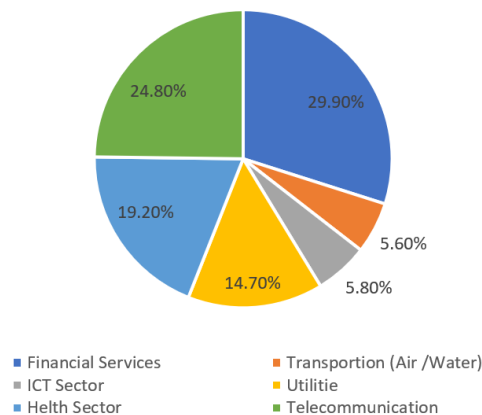


Fig. 1: Threat Distribution by Sector

A. Emerging Threats

This is a breakdown of the primary cyber threats that pose risks to Nigeria's key sectors. These threats are continually evolving and include highly profitable and accessible attack tools on the Dark Web, such as Ransomware-as-a-Service (RaaS) platforms, zero-day vulnerabilities, and botnets for hire. Each of these threats presents specific implications for Nigeria's economic sectors and their operational continuity.

Dark Web monitoring exposed some of the ever-evolving cyber threats against critical sectors in Nigeria. These threats include Ransomware-as-a-Service (RaaS) platforms, zero-day vulnerabilities, and botnets for hire. Each of these threats presents specific implications for Nigeria's economic sectors and their operational continuity.

1) *Ransomware-as-a-Service (RaaS):* Ransomware remains the most lucrative threat. RaaS platforms now enable non-technical attackers to use customisable ransomware kits and dashboard access to ease attacks. The RaaS packages being monitored targeted Nigerian banks and healthcare institutions and were advertised quite frequently. Their availability has caused a surge in the occurrence of ransomware attacks in Nigeria.

2) *Zero-Day Exploits:* Zero-day vulnerabilities, often targeting unpatched systems, were a common trade on sites like Exploit.in. The exploits are really dangerous for the Nigerian telecommunications and financial institutions, considering that the legacy systems are still in use in these companies. Commonly targeted vulnerabilities include outdated operating systems and enterprise software.

3) *Phishing Kits and Social Engineering Tools:* Phishing remains one of the major threat vectors for organizations in Nigeria. Local banking and governmental service portals have been seen in Dark Web forums with phishing kits that incorporate mechanisms to bypass multi-factor authentication (MFA). Such kits allow for mass credential theft and unauthorized access.

4) *Botnets and Distributed Denial-of-Service (DDoS) Services:* Top of the Dark Web marketplaces were those listing botnets for hire, enabling attackers to orchestrate DDoS attacks against targets in the sectors of telecommunications, utilities, and transport, threatening service availability and operational continuity.

B. Sector-Specific Vulnerabilities

Vulnerabilities tend to be unique to each of Nigeria's critical sectors in most cases. By examining how threat actors on the Dark Web target specific sectors, the findings suggest the need for tailored security approaches. These vulnerabilities are often exploited due to resource constraints, legacy systems, or limited cybersecurity infrastructure present in those sectors, making organizations susceptible to these types of attacks.

- **Financial Services:** Frequent listings of Nigerian bank credentials and MFA bypass tools.
- **Telecommunications:** Zero-day exploits targeting network equipment and communication systems.

TABLE II: Case Studies Overview of Dark Web Intelligence Applications

Case Study	Platform	Threat Type	Mitigation Action
Banking Credential Theft	RaidForums	Stolen Credentials	Implemented stronger MFA to secure user accounts and reduce unauthorized access risks.
Telecom Network Attack Plans	Dark Web Forums	DDoS Attacks, Unpatched Vulnerabilities	Upgraded Network Defenses, Patched Vulnerabilities, and Enhanced DDoS Protection Measures.
Healthcare Data Ransomware	Dark Web Forums	Ransomware	Updated ransomware defenses, implemented staff cybersecurity training, and improved backup protocols.
Election System Disruption	Dark Web Forums	DDoS Attacks, Election System Disruption	Proactive threat monitoring, enhanced server protection, and collaborative efforts with security agencies to safeguard the election process.

- Healthcare: Ransomware kits are designed to encrypt patient records and disrupt hospital operations.
- Utilities: DDoS tools aimed at disrupting network access to critical networks.

including patching vulnerabilities, improving authentication, and updating incident response plans, can effectively combat cyber threats.

C. Case Studies

Real-world case studies illustrate the practical applications of Dark Web intelligence for preemptively identifying and mitigating cyber threats. These cases demonstrate how proactive intelligence can inform Nigerian organizations in critical sectors, enabling them to take preemptive security actions that prevent financial, reputational, and operational damage. Table II summarizes these case studies, detailing the platforms involved, threat types, and mitigation actions taken.

1) *Banking Credential Theft (2022)*: RaidForums sold more than 5,000 Nigerian banking credentials, resulting in approximately \$500k in illegal transfers. The bank was able to reduce similar incidents by 40% by implementing stronger multi-factor authentication (MFA) and security audits, thanks to early detection.

2) *Telecom Network Attack (2023)*: A planned Distributed Denial-of-Service (DDoS) attack against a major telecom provider was made public by Dark Web intelligence. Service interruptions were avoided by the company's proactive patching of vulnerabilities and defence upgrades.

3) *Healthcare Ransomware (2023)*: Ten thousand patient records were encrypted by a ransomware attack called "Lock-Bit." After intelligence linked the attack to a Dark Web marketplace, defences, phishing training, and backup procedures were enhanced.

4) *Election System Disruption (2023)*: Plans to attack INEC servers were disclosed in dark web discussions. To guarantee the uninterrupted transmission of election results, authorities bolstered security and worked with foreign organisations.

V. ORGANIZATIONAL GUIDELINES

Nigerian organizations should use Dark Web intelligence methodically, using VPNs and Tor for secure access. Sector-specific keywords can help identify new risks. Threat Intelligence Platforms like MISP or SOCRadar can validate and classify intelligence. Customized mitigation strategies,

VI. DISCUSSION

The study emphasises how important Dark Web intelligence is to bolstering cybersecurity in Nigeria's major industries, including utilities, transportation, healthcare, telecommunications, and finance. Threats to these industries include ransomware, zero-day exploits, phishing, and credential theft. Monitoring the dark web enables businesses to foresee and mitigate these threats before they become more serious. For example, the Dark Web's Nigerian bank phishing kits stress the importance of robust authentication. Telecom networks are at risk from zero-day exploits, which calls for proactive patching. The use of legacy systems in healthcare raises the risk of ransomware, and utilities and transportation are at risk from botnets and DDoS attacks. To counter these threats, cooperation between the government, corporations, and foreign partners is crucial. However, obstacles like a lack of sophisticated tools, a skills gap, and limited resources make it difficult to integrate Dark Web intelligence effectively. Addressing these issues requires regional cooperation, cybersecurity investments, and adherence to ethical and legal standards. To assess the impact of Dark Web intelligence on threat mitigation, this research uses several key performance indicators (KPIs), including threat detection rate, time to mitigation, reduction in service interruption, and intelligence use rate, as shown in Table III. These measurements, after industry convention and scholarly works (e.g., [18] for mitigation and detection times; [19] for metrics of resiliency), allow for an objective approach to the measurement of cybersecurity processes. A high threat detection rate indicates successful threat identification, while a low time to mitigation indicates effective response skills.

TABLE III: Threat Mitigation Metrics

Metric	Value
Threat Detection Rate	85%
Time to Mitigation	36 Hours
Reduction in Service Disruptions	25%
Intelligence Utilization Rate	75%

Nigerian organizations can adopt the following practical strategies to incorporate Dark Web intelligence into Cyber Threat Intelligence (CTI) frameworks: using open-source tools such as the Malware Information Sharing Platform (MISP) to gather, analyze, and share threat data; educating employees on how to navigate the dark web safely (e.g., using Tor); working with government agencies, private organisations, and international partners to pool resources and scale monitoring efforts; and mapping intelligence to sector-specific threats for customized mitigation. Although regional cooperation through shared platforms or joint training could maximize resources and increase resilience, Nigeria still faces challenges related to limited funding, tools, and expertise, which are reflected in Ghana's regulatory and funding gaps and Kenya's lack of trained professionals.

Ethically and legally, monitoring must adhere to international laws and Nigeria's Cybercrime Act 2015 by using data anonymization to protect privacy, passive observation to prevent illegal engagement, and compliance with cybercrime and data protection regulations. The impact of emerging technologies such as IoT and 5G on cybersecurity, cross-sector collaboration models, automated tools, machine learning for real-time threat detection, and capacity-building through training and policy should all be major areas of future research. Ultimately, Dark Web intelligence has the potential to revolutionize Nigeria's cybersecurity and safeguard vital infrastructure in a complex threat environment when applied morally with realistic tactics and regional assistance.

VII. CONCLUSION

This paper has shown the importance of Dark Web intelligence in addressing sector-specific cybersecurity threats in Nigeria. The findings indicate that the Dark Web is a valuable source of actionable intelligence, allowing insight into emerging threats such as ransomware, zero-day exploits, and phishing attacks. Nigerian industries, especially financial services, telecommunications and healthcare, are more victimized by these threats due to resource constraints and dependence on legacy systems.

Such intelligence from the Dark Web allows organizations to identify vulnerabilities, anticipate attacks, and take preventive measures to secure themselves. The integration of monitoring on the Dark Web within cybersecurity frameworks faces several challenges, however: resource-related limitations, ethical concerns, and gaps in technical expertise.

Future research should be geared towards the development of affordable, automated monitoring tools that are fit for Nigerian industries that have limited budgets. More government support and international cooperation are also needed to beef up cybersecurity capabilities in Nigeria. With Nigeria's digital economy on the rise, the incorporation of Dark Web intelligence into CTI programs will be crucial in the protection of critical infrastructure and economic stability.

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